ARGUS

Autonomous Real-time Gateway Unsupervised Shield

AI-POWERED REAL-TIME NETWORK SECURITY

In today's connected world, security is not an option — it's a necessity. Let's safeguard the digital age together.

THE GROWING DIGITAL SECURITY GAP:

As internet connectivity expands across homes, schools, hospitals, and rural communities, cybersecurity protection has not grown at the same pace.

Challenges:
- Weak router security
- No active monitoring
- Limited cybersecurity awareness
- Lack of dedicated security personnel

Consequences:
- Data theft
- Privacy breaches
- Financial loss
- Service disruption

THE PROTECTION GAP:

Traditional Firewalls

- Static rules
- Limited visibility
- Misbehavioral anomalies

Enterprise IDS/IPS

- Powerful
- Expensive
- Complex deployment requiring experts

Small organizations need enterprise-level protection without enterprise-level complexity.

INTRODUCING ARGUS:

ARGUS is an AI-powered Network Anomaly Detection and Blocking System designed to provide intelligent cybersecurity protection for local networks.

Capabilities:

- Monitor network traffic
- Detect suspicious activity
- Classify threats
- Alert administrators
- Block confirmed attackers
- Visualize incidents in real time

Mission:

Making advanced cybersecurity accessible to every connected community.

VISUAL FLOW:

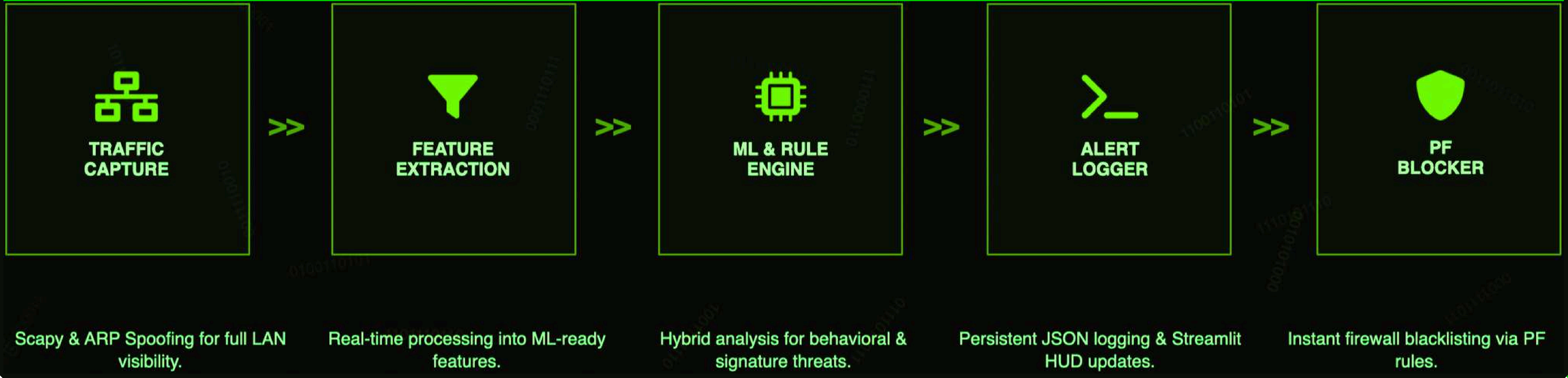
Network Traffic → Traffic Analysis → Feature Extraction → Threat Detection → Classification → Blocking → Dashboard

ARGUS continuously observes network behavior and identifies threats before they escalate.

COMPONENTS:

Traffic Visibility Layer, Packet Capture Engine, Feature Extraction Module, Machine Learning Engine, Rule-Based Detection Engine, Decision Layer, Blocking Engine, Dashboard Layer

Traffic is captured and analyzed in real time. Threat intelligence is generated using both ML and explicit signatures.



HYBRID THREAT DETECTION:

Rule-Based Detection

Detects:

- Port scans
- Host sweeps
- SYN floods

Machine Learning Detection

Isolation Forest Model

Detects:

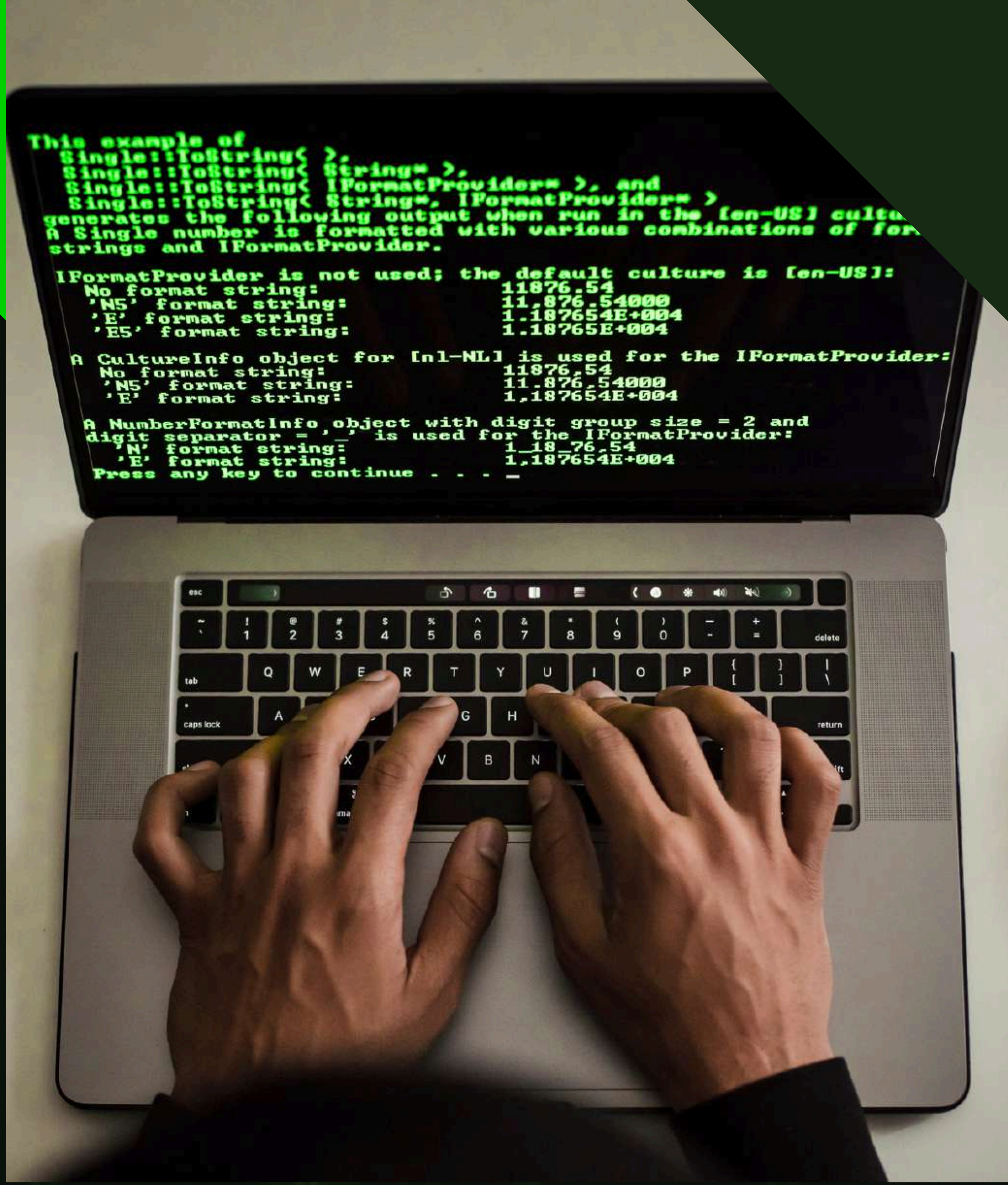
- Traffic anomalies
- Behavioral deviations
- Unknown attack patterns

Key Innovation:

Known attacks are blocked immediately, while suspicious, unknown behavior is intelligently analyzed.

THREATS DETECTED INCLUDE:

- TCP Port Scans
- SYN Scans
- Connect Scans
- Host Sweeps
- Reconnaissance Activity
- SYN Flood Attacks
- Statistical Traffic Anomalies



TECHNICAL APPROACH:

Programming Language:
Python

Packet Analysis:
Scapy

Machine Learning:
scikit-learn

Model:
Isolation Forest

Data Processing:
Pandas, NumPy

Visualization:
Streamlit

Training Datasets:

- NSL-KDD
- KDDTest
- CICIDS2017

REAL-TIME DASHBOARD:

Features:

- Live alert feed
- Threat severity scoring
- Attack classification
- Timeline analysis
- Source investigation
- Historical logs

REAL-TIME DASHBOARD:

OPERATIONAL WORKFLOW:

Step 1:
Traffic Monitoring

Step 2:
Packet Capture

Step 3:
Feature Extraction

Step 4:
ML + Rule Evaluation

Step 5:
Threat Classification

Step 6:
Alert Generation

Step 7:
Blocking Decision

Step 8:
Dashboard Update

Attacker → Traffic → Detection → Classification → Alert→ Block → Protected Network

SAFEGUARDS:

- Trusted device whitelist
- Monitor mode
- Configurable thresholds
- Reduced false positives
- Safe shutdown
- Controlled automation

Important Statement:
Machine-learning anomalies alone are not automatically blocked without additional validation.

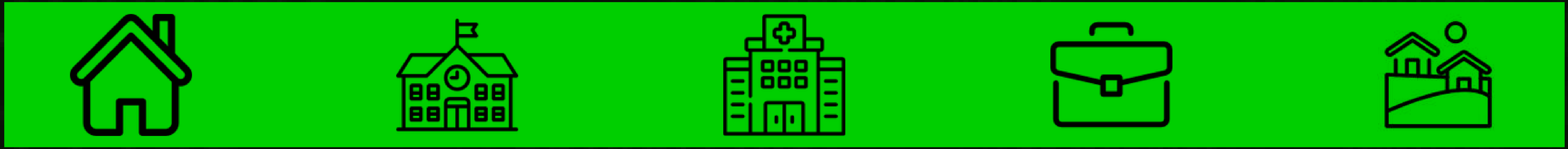
IMPACT & BENEFICIARIES:

Beneficiaries:

- Homes
- Schools
- Hospitals
- Small Businesses
- Rural Digital Centers

Benefits:

- Improved cybersecurity access
- Reduced cybercrime risk
- Early attack detection
- Increased trust in digital systems



SCALABILITY & FUTURE ROADMAP:

Phase 1:

Current ARGUS

Phase 2:

• Telegram Alerts

• Slack Alerts

•Threat Intelligence

Phase 3:

• Web Portal

• Mobile App

• Multi-Network Monitoring

Phase 4:

• Edge AI Security Appliance

• Federated Learning

• Enterprise Analytics

PROJECT VISION:

Cybersecurity should not be a privilege reserved for large organizations.
ARGUS aims to democratize intelligent network protection and make advanced security accessible to every connected community.

FUTURE VISION:

Affordable AI-driven cybersecurity for everyone.

CONCLUSION:

ARGUS combines:

• Artificial Intelligence
• Behavioral Analysis
• Real-Time Detection
• Automated Protection

to provide practical cybersecurity for underserved networks.

THANK YOU!

Thank you for being part of the secure digital journey.